

Project Design Phase-II

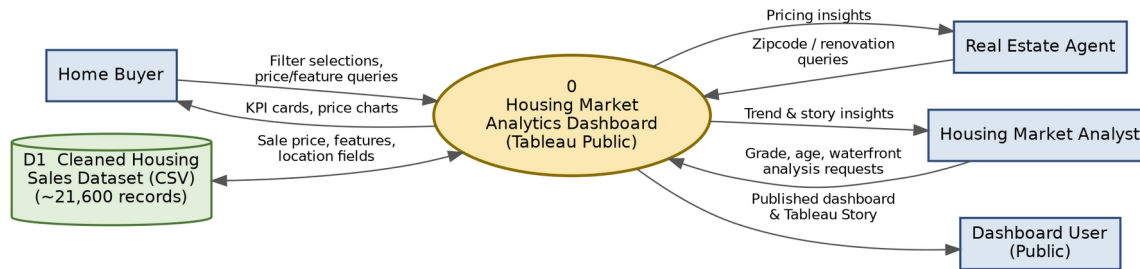
Data Flow Diagram & User Stories

Date	13 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

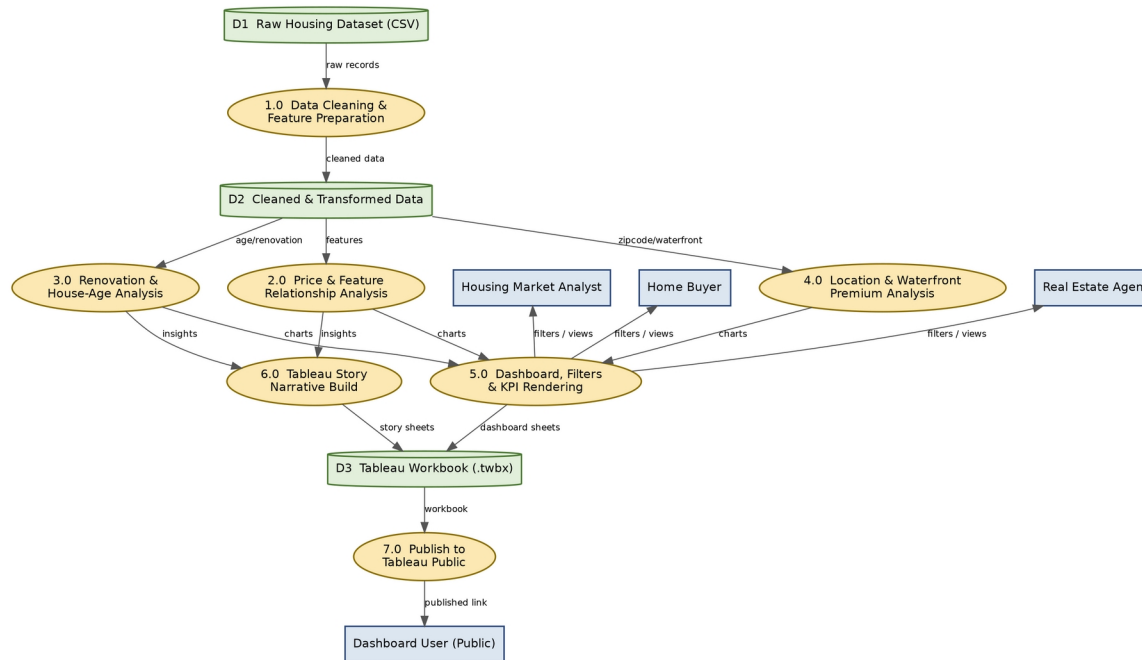
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A clear DFD depicts the right amount of the system requirement graphically — how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 — Context Diagram



DFD Level 1 — Process Breakdown



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	Story #	User Story / Task	Acceptance Criteria	Priority	Release
Home Buyer	KPI Dashboard	USN-1	As a Home Buyer, I want to view key KPIs such as total properties, average sale price, and total living area so that I can quickly gauge overall market conditions.	KPI cards display accurate, up-to-date totals when the dashboard loads.	High	Sprint-1
	Interactive Filters	USN-2	As a Home Buyer, I want to filter listings by	Selecting a filter updates all visible	High	Sprint-1

			zipcode group and price range so that I can focus on homes within my budget and preferred location.	charts and KPIs without a page reload.		
	Price & Feature Analysis	USN-3	As a Home Buyer, I want to see how sale price varies with bedrooms and bathrooms so that I can judge whether a listing is fairly priced.	Price-by-feature chart updates correctly when filters are applied.	Medium	Sprint-2
Real Estate Agent	Renovation Impact Analysis	USN-4	As a Real Estate Agent, I want to analyse how renovation status and years since renovation affect sale price so that I can advise clients on renovation-driven pricing.	Renovation chart shows an accurate breakdown of sale counts / prices by renovation status.	High	Sprint-2
	Location-wise Price Analysis	USN-5	As a Real Estate Agent, I want to compare price trends across zipcode groups so that I can recommend competitive listing prices to sellers.	Zipcode-wise chart reflects average / total sale price matching the underlying dataset.	Medium	Sprint-2
	House Age Analysis	USN-6	As a Real Estate Agent, I want to view house-age distribution by bedrooms, bathrooms and floors so that I can identify which configurations are common in older vs. newer homes.	Age-distribution chart correctly facets by bedrooms, bathrooms and floors.	Medium	Sprint-3
Housing Market Analyst	Price & Feature Relationship Analysis	USN-7	As a Housing Market Analyst, I want to explore the relationship between overall grade / condition and sale price so that I can quantify how quality drives value.	Grade vs. price chart shows a clear, correctly aggregated relationship.	High	Sprint-1
	Waterfront & View Premium Analysis	USN-8	As a Housing Market Analyst, I want to analyse the waterfront-view premium so that I can measure its impact on sale price.	Waterfront comparison view shows average price with and without a waterfront view.	Medium	Sprint-3
	Tableau Story	USN-9	As a Housing Market Analyst, I want to walk through a guided Tableau Story summarising key housing findings so that I can present insights without rebuilding charts.	Story points navigate in sequence and each point displays its correct supporting visualization.	Medium	Sprint-3
	Renovation Impact Analysis	USN-10	As a Housing Market Analyst, I want to track total sales by years since renovation so that I can identify how recency of renovation influences market activity.	Sales-by-renovation chart displays correct totals for each renovation-year bucket.	High	Sprint-1
	Interactive Filters	USN-11	As a Housing Market Analyst, I want interactive filters for bedrooms, bathrooms and zipcode group so that I can drill into specific market segments.	Applying any filter updates all dependent charts and KPI cards consistently.	High	Sprint-1
Dashboard User	Public Access & Sharing	USN-12	As a Dashboard User, I want to access the published Tableau dashboard and story directly	The Tableau Public page loads the dashboard and story correctly in	High	Sprint-1

			via Tableau Public so that I can view insights without installing any software.	common browsers.		
	Dashboard Visualization	USN-13	As a Dashboard User, I want the dashboard layout to be clear and easy to navigate so that I can find the information I need without confusion.	Navigation between dashboard sections (KPI, Price, Renovation, Age, etc.) is available from a single visible layout.	Medium	Sprint-2
	Interactive Filters	USN-14	As a Dashboard User, I want to reset all filters with a single action so that I can quickly return to the default dashboard view.	A Reset Filters control clears all active filters and restores the default view.	Low	Sprint-2